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Digest paper

Recent topics on catalytic asymmetric 1,4-addition



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ABSTRACT

Catalytic asymmetric 1,4-addition (conjugate addition; Michael addition) is one of the most powerful methods for carbon-carbon bond formation. Following the first efficient catalyst system developed by Feringa, which is composed of $\text{Cu}(\text{OTf})_2$ and phosphoramidite with dialkylzincs, a variety of chiral catalysts have been reported for the catalytic asymmetric conjugate addition. In this digest review, we will first summarize novel chiral ligands that work efficiently for cyclic and acyclic enones and demonstrate the wide applicability of Michael acceptors. We will also introduce unique phenomena that include the nonlinear effect and reversal of enantioselectivity. Organomagnesium reagents have also been used instead of organozincs. Finally, we introduce the recent examples of the synthesis of natural products based on the catalytic asymmetric reaction. The rare experimental studies into the mechanism of copper-catalyzed 1,4-addition reported by Kitamura and Noyori's group are also introduced.

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Introduction

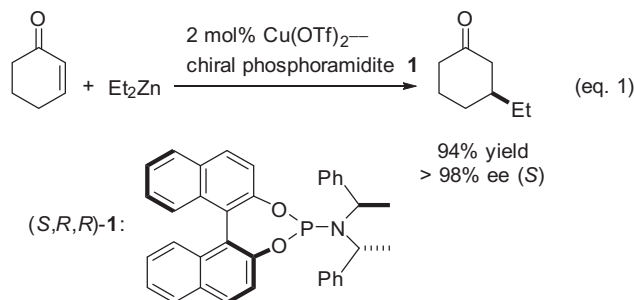
Catalytic asymmetric 1,4-addition is now recognized as one of the most reliable carbon-carbon bond forming reactions in organic synthesis.^{1,2} In 1988, Soai and his co-workers first reported the asymmetric 1,4-addition of chalcone with diethylzinc in the presence of $\text{Ni}(\text{acac})_2$ and chiral β -amino alcohol to furnish the

1,4-addition product in 75% yield with 45% ee.³ Subsequently, there have been many reports on asymmetric 1,4-addition reactions, most of which included the reactions of cyclic and acyclic enones with dialkylzincs in the presence of chiral copper complexes. A remarkable improvement was introduced by the combination of $\text{Cu}(\text{OTf})_2$ and chiral phosphoramidite **1** developed by Feringa in 1997 (Eq. 1).^{4,5}

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Feringa (1997)



A general and rational mechanism underlying the copper-catalyzed 1,4-addition of α,β -unsaturated ketones using dialkylzincs and a $\text{Cu}(\text{OTf})_2$ -chiral ligand system is outlined in Scheme 1.^{4,6,7} However, since it is not easy to isolate and analyze the reaction intermediates, the actual reaction mechanism is still under debate.

In this article, we present an overview of the recent progress of catalytic asymmetric 1,4-addition.

Development of effective chiral ligands

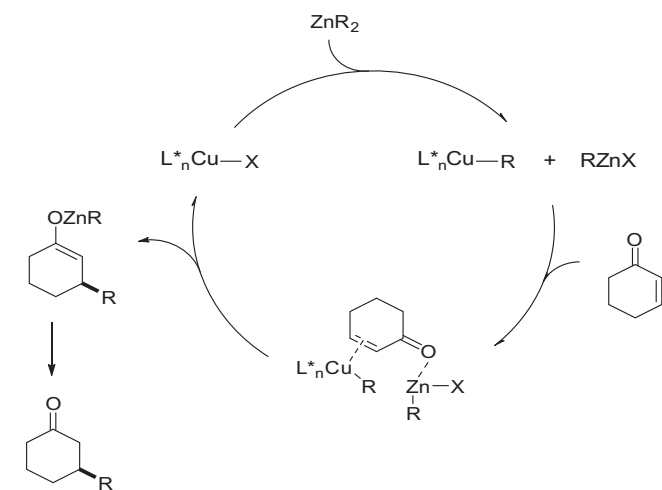
Following Feringa's report,⁴ continuous efforts have been undertaken for the development of new chiral ligands that afford high reactivity and high enantioselectivity. In this regard, various binaphthol-derived phosphoramidites, phosphonites, and phosphites were reported. Some of the ligands that gave the 1,4-addition product in the reaction of 2-cyclohexen-1-one in >95% ee are summarized in Scheme 2.^{8–12}

Other types of ligands that afford a high ee include chiral Schiff bases. The successful examples of chiral ligands for conjugate addition are shown in Scheme 3.^{13–15}

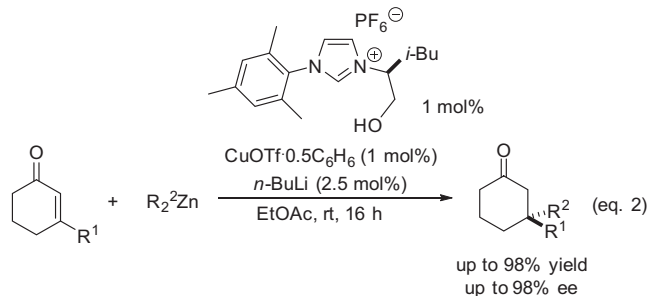
In 2001, Fraser and Woodward reported a strong ligand acceleration effect upon the addition of *N*-heterocyclic carbene (NHC) in the reaction of various enones with diethylzinc, and proposed the transition state illustrated in Scheme 4.¹⁶

Based on this concept, Mauduit developed several chiral alkoxy NHC ligands (derived from alcohol and *n*-BuLi) (Scheme 5).^{17–19}

The same authors recently reported the reaction of β -substituted cyclohexenones with dialkylzincs to afford products possessing quaternary carbon centers, in up to 98% ee (Eq. 2).²⁰

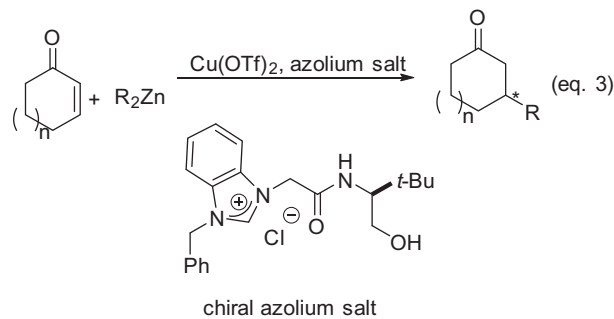


Scheme 1. Proposed mechanism for the copper-catalyzed 1,4-addition of organozincs to 2-cyclohexenone.



Sakaguchi and his co-workers developed and reported a similar type of ligand, as shown below (Eq. 3). In this case, the chiral azolium salts themselves were found to work as catalysts. In addition, this group observed a unique phenomenon, reversal of enantioselectivity which will be discussed later.²¹

Sakaguchi (2010)



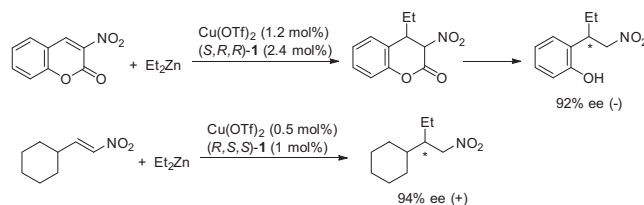
For acyclic enones, the chiral ligands shown below exhibited high enantioselectivity (Scheme 6).^{22–33}

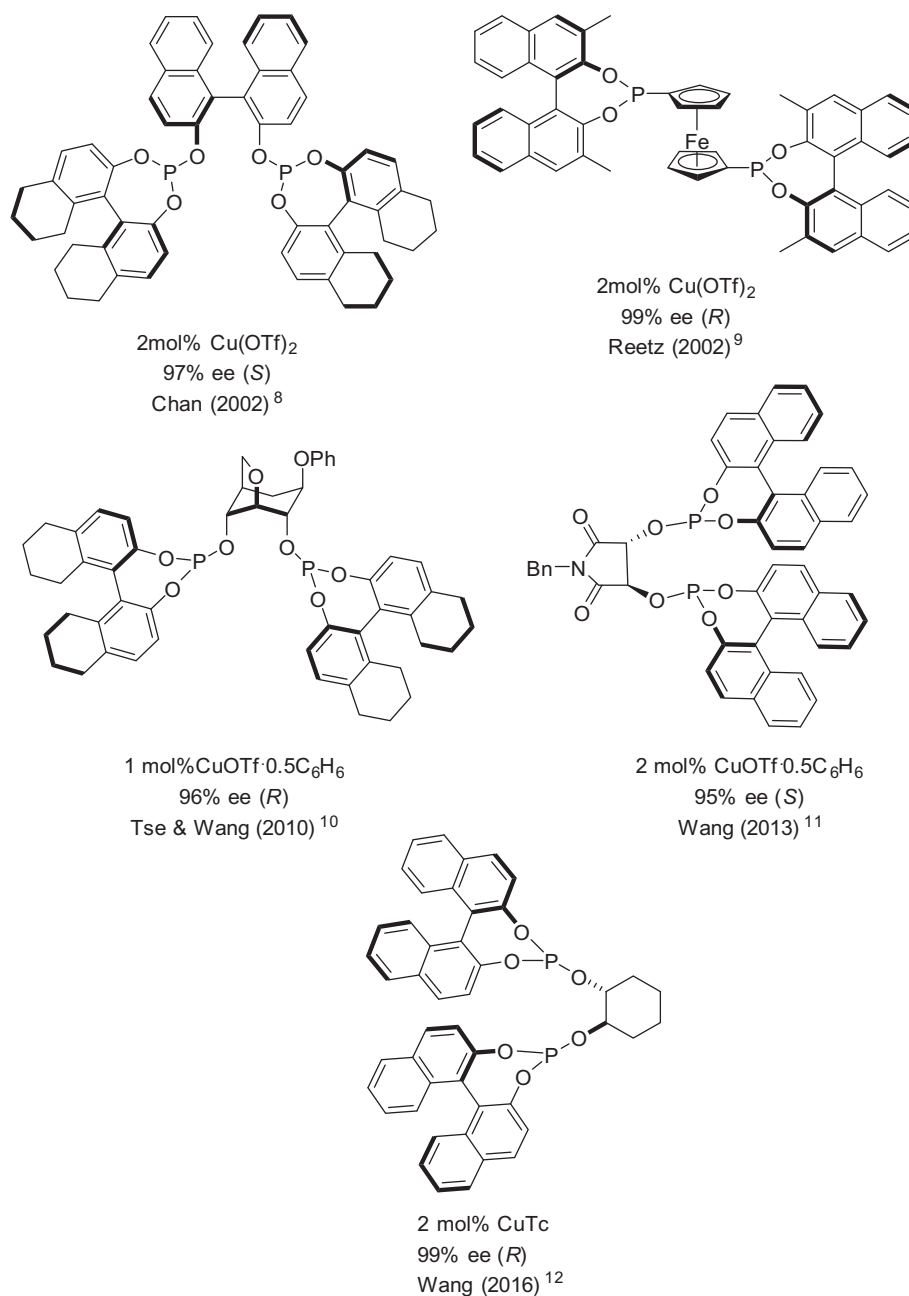
Expansion of scope of Michael acceptors

In most of the catalytic asymmetric 1,4-additions reported so far, cyclic and acyclic cyclohexenones such as 2-cyclohexen-1-one and chalcone, respectively, were used as substrates. In this chapter, we will introduce various Michael acceptors other than the abovementioned simple enones. In 1999, Feringa reported the copper-phosphoramidite-catalyzed 1,4-addition of dialkylzincs to symmetric 4,4-disubstituted cyclohexadienones and unsymmetric 4,4-disubstituted cyclohexadienones to give the 1,4-addition products in high ee (Scheme 7).³⁴

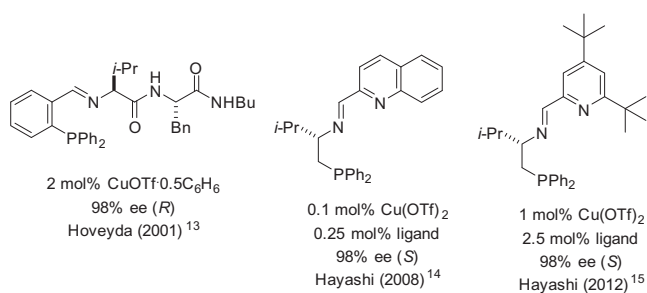
Feringa extended this methodology for the synthesis of both *cis*- and *trans*-3,4,4,5-tetrasubstituted cyclohexanones by appropriate choice of (*S,R,R*)-1 or (*R,S,S*)-1. The reaction proceeded selectively with chiral catalyst control (Scheme 8).³⁵

As Michael acceptors, nitroolefins were also used to give the corresponding nitroalkanes; however, only two substrates have been reported to give the product with high ee.^{36,37} Feringa reported the asymmetric conjugate addition of dialkylzinc reagents to α,β -unsaturated nitroacetates, which led to the enantioselective synthesis of β -aryl-nitroalkanes.



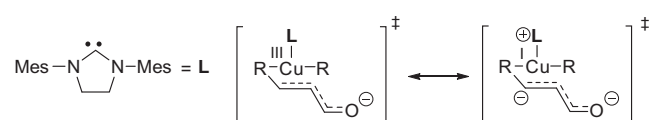


Scheme 2. Binaphthol-derived ligands for asymmetric 1,4-addition.

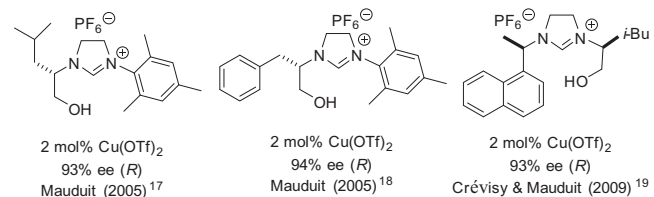


Scheme 3. Chiral Schiff base ligands for asymmetric conjugate addition.

Recently, Mauduit, and Campagne reported the copper-catalyzed enantioselective conjugate addition of dimethylzinc to unsaturated 2-acyl-*N*-methylimidazole using a chiral bidentate

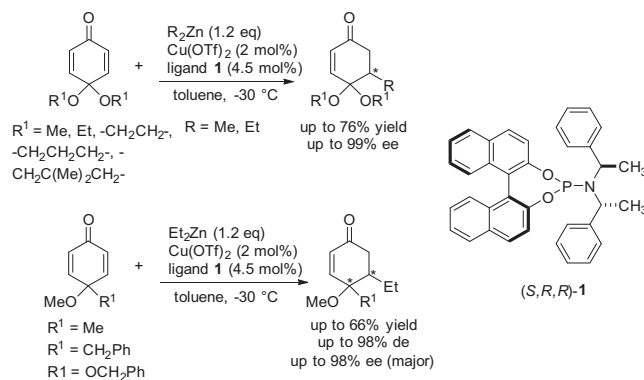
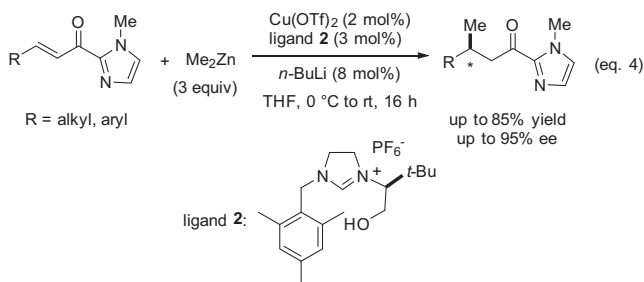


Scheme 4. Mechanism of ligand accelerated catalysis in 1,4-addition.

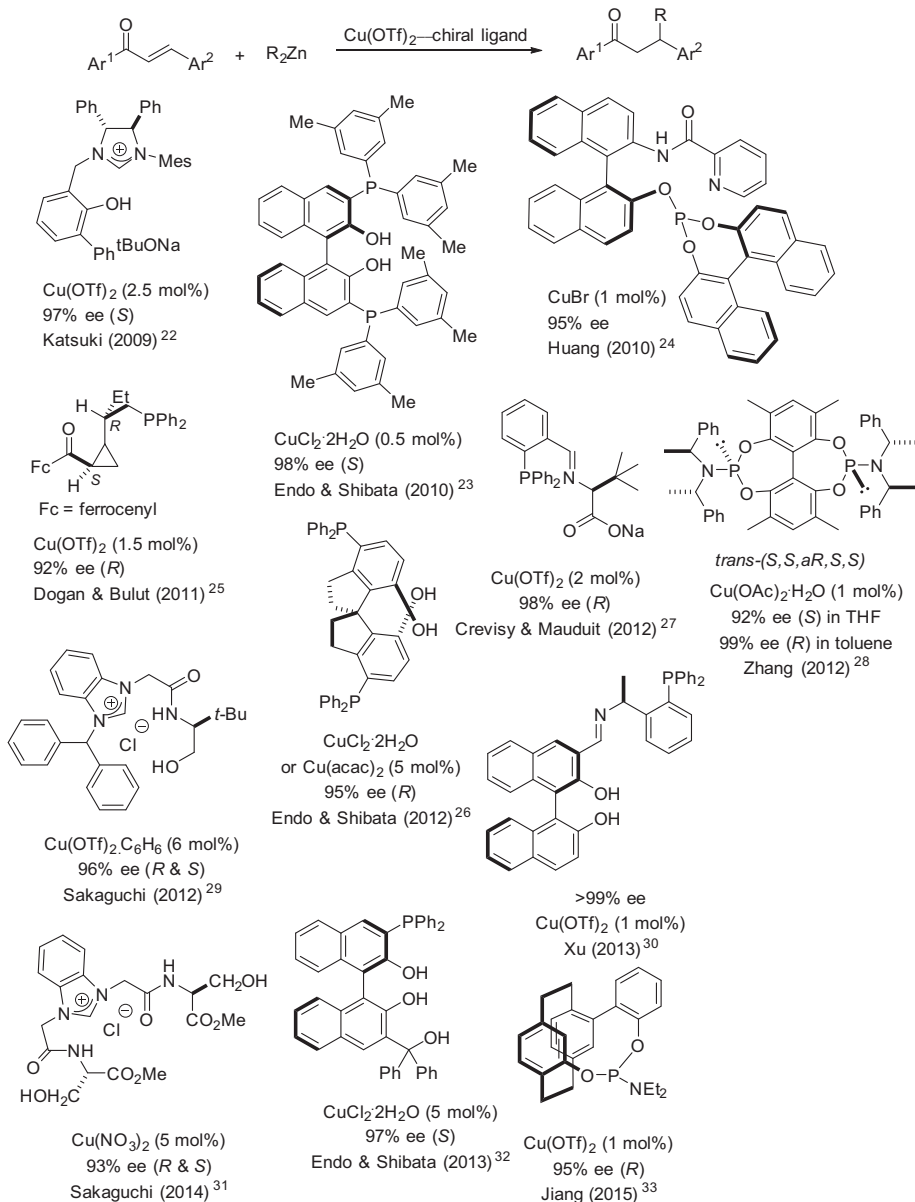


Scheme 5. Chiral NHC ligands for asymmetric 1,4-addition.

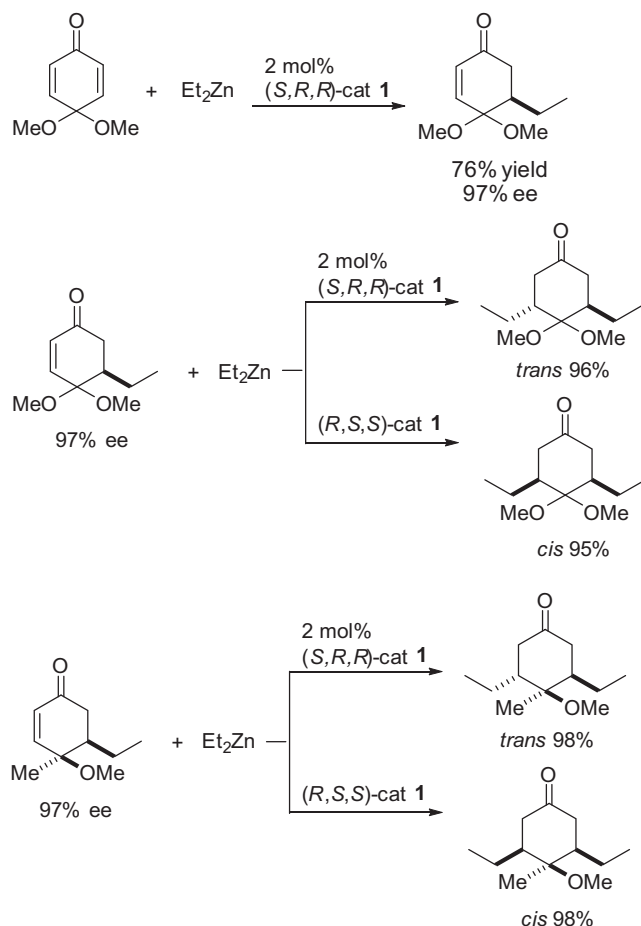
hydroxyalkyl-NHC ligand (Eq. 4).³⁸ The 1,4-addition product, acylimidazole, was further converted to the corresponding aldehyde by treatment first with NaBH₄, and then with MeI and 2 M NaOH.



Scheme 7. Asymmetric conjugate addition of symmetric and unsymmetric 4,4-disubstituted cyclohexadienones.



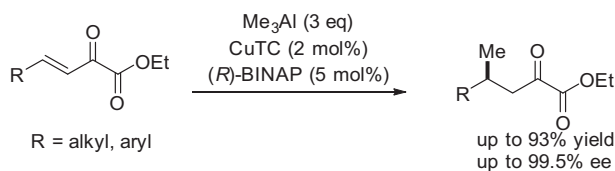
Scheme 6. Efficient chiral ligands for acyclic enones.



Scheme 8. Synthesis of *cis*- and *trans*-3,4,4,5-tetrasubstituted cyclohexanone.

The unsaturated acyl imidazole obtained after the Horner-Wadsworth-Emmons reaction was then employed for the asymmetric 1,4-addition to give the desoxypropionate unit (Scheme 9). As an application of this protocol, the authors achieved the enantioselective synthesis of ionone derivatives and (+)-*ar*-turmerone (Scheme 10).

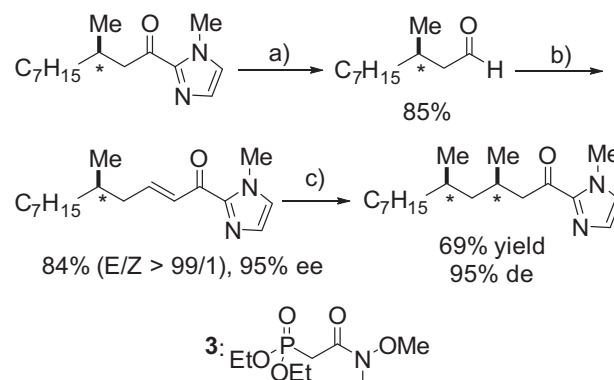
The same authors also reported a DFT study on the mechanism underlying the regioselective formation of the 1,4-addition product over the 1,6-addition product.³⁹ Alexakis reported the asymmetric 1,4-addition of β,γ -unsaturated α -ketoesters using trimethylaluminum instead of dialkylzinc.⁴⁰



Using this catalyst system, the authors achieved the efficient synthesis of (*S*)-florhydral (Scheme 11) and unnatural α -amino acid precursors (Scheme 12).

Trapping of zinc enolates by some electrophiles

One of the characteristic features of copper-catalyzed 1,4-addition using dialkyl zincs is the generation of an intermediate zinc enolate. Therefore, if this zinc enolate is trapped by an electrophile,



a) 1) NaBH_4 , MeOH, rt, 2 h; 2) MeI, EtOAc, 60 °C, 16 h; 3) 2 M NaOH, glycine, toluene, 80 °C, 5 h. b) 1) **3**, NaH, THF, -78 °C to rt, 16 h; 2) NMI, *n*-BuLi, THF, -78 °C to rt, 5 h. c) Me_2Zn (3 equiv), $\text{Cu}(\text{OTf})_2$ (2 mol%), ligand **2** (3 mol%), *n*-BuLi (8 mol%), THF, 0 °C to rt, 16 h.

Scheme 9. Synthesis of desoxypropionate unit.

multicomponent coupling can be realized. In their first report of asymmetric 1,4-addition in 1997, Feringa et al. described the reaction of a zinc enolate with aldehydes (Scheme 13).^{4b}

In 1999, Feringa also reported catalytic enantioselective annulations via 1,4-addition and aldol condensation to afford $[n + 4]$ and $[n + 3]$ annulation products. They reported three-component coupling using an allyl acetate and a catalytic amount of $\text{Pd}(\text{PPh}_3)_4$. A typical example of $[n + 3]$ annulation is shown in Scheme 14.⁴¹

Alexakis and his-coworkers reported the reaction of zinc enolate with activated electrophiles such as nitroolefins and vinylsulfonates in the absence of a Pd catalyst (Schemes 15 and 16).^{42,43}

We also investigated the direct trapping of the intermediate zinc enolates with allyl iodides. The results are summarized in Table 1. The reactions were carried out using 1 mol% of $\text{Cu}(\text{OTf})_2$ and 2.5 mol% of the ligand.⁷ Table 2

Alexakis also reported the reaction of zinc or aluminum enolate with vinyl oxiranes, which led to the formation of optically active allylic alcohols with up to 98% ee. This methodology was applied to the formal synthesis of *Clavularin B* (Scheme 17).⁴⁴

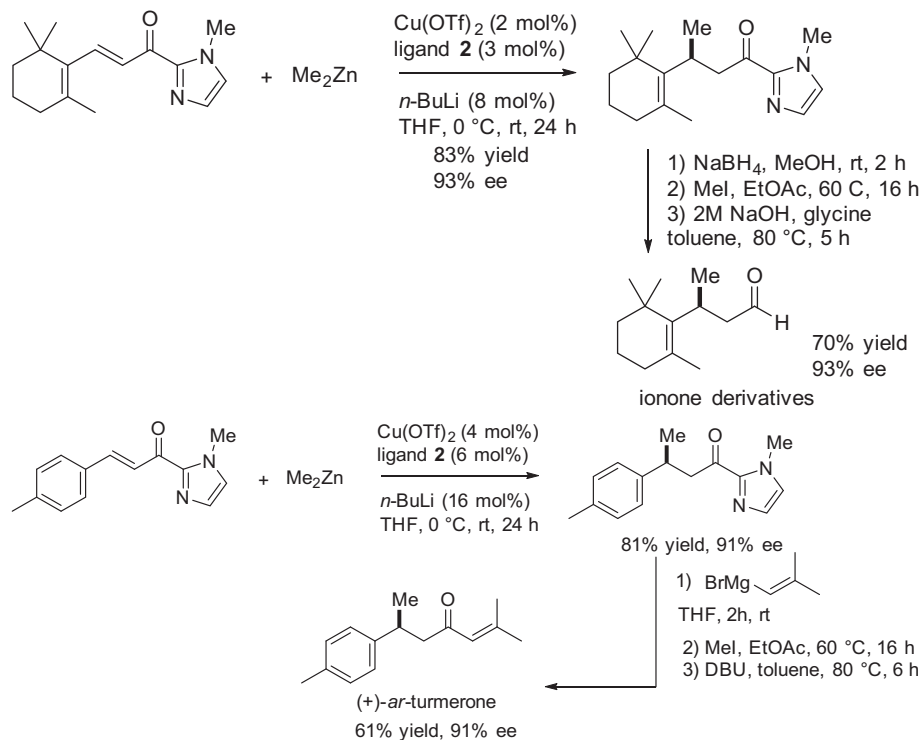
Huang reported a diastereo- and enantioselective dual tandem reaction that affords three contiguous stereocenters and disclosed the synthesis of functionalized pyrrolidine (Scheme 18).⁴⁵

Experiment-based investigation of reaction mechanism

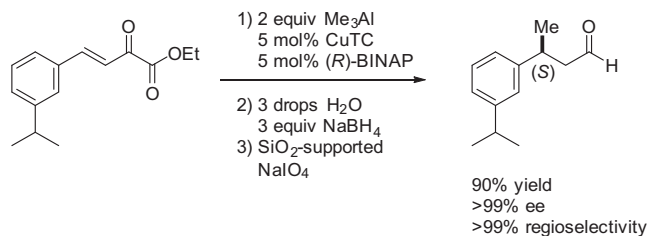
As mentioned at the beginning of this article, detailed studies on the mechanism of 1,4-conjugate addition are very rare. As an exception, Kitamura and Noyori carefully elucidated the reaction mechanism for a dialkylzinc and copper(I)-sulfonamide system based on a kinetic study and structural analysis of the zinc enolate product by NMR and molecular weight measurements (Scheme 19).⁴⁶ They proposed the mixed metal complex **A** as an active species. Complex **A** generates the catalyst/reagent/substrate complex **B** in an unfavorable equilibrium. The turnover limiting step is the following irreversible alkylation step, where alkyl group transfer is facile due to the sulfonamide spacer. The formation of stable dimeric alkylzinc enolate **C** is proposed to facilitate the regeneration of active species **A**.

Non-linear effect

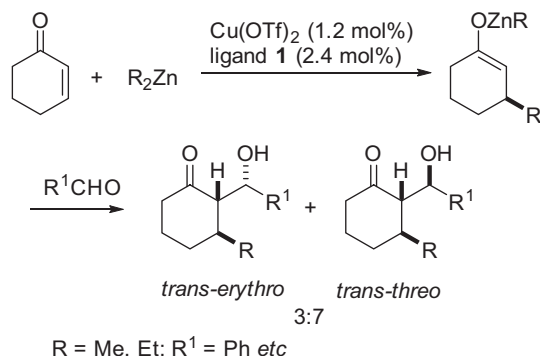
Following the first discovery of the non-linear effect in catalytic asymmetric reactions by Kagan in 1986,⁴⁷ numerous other reports



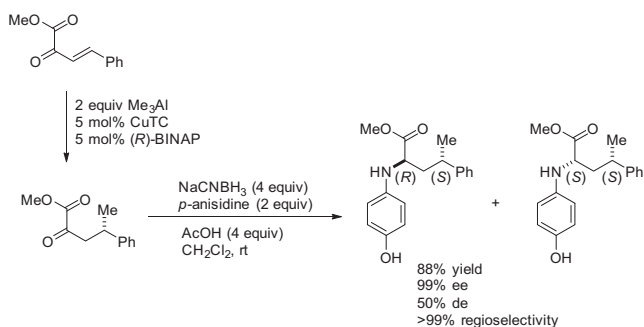
Scheme 10. Synthesis of (+)-ar-turmerone.



Scheme 11. Synthesis of (S)-florhydral.



Scheme 13. Tandem 1,4-addition-aldol reaction.



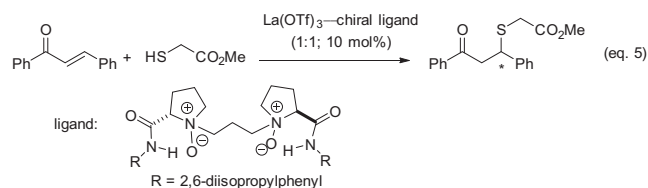
Scheme 12. Synthesis of unnatural α-amino acid precursors.

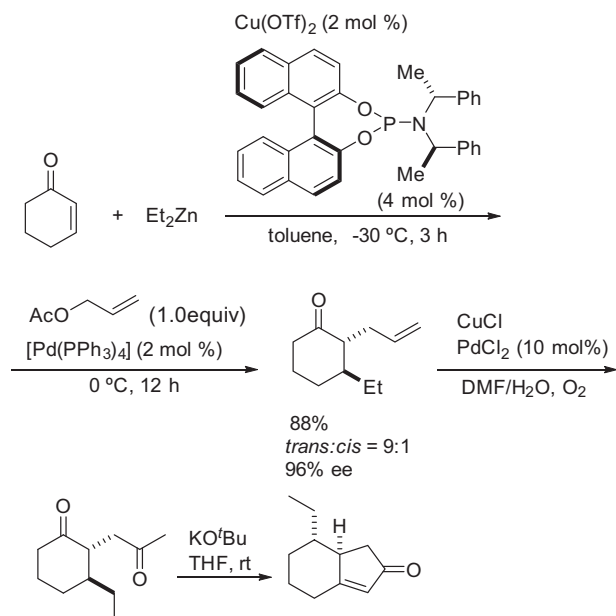
have discussed this phenomenon.⁴⁸ In the asymmetric 1,4-addition carried out using a copper catalyst and dialkylzinc system, both positive and negative nonlinear effects were observed, which often give us useful information concerning the intermediate complexes, especially in the aggregation state.

Feringa's group also contributed to information regarding the non-linear effect in 1,4-addition, from an early stage. They

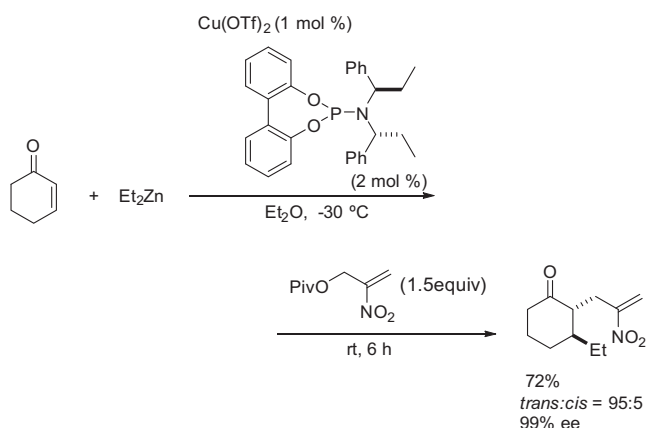
reported a positive non-linear effect (asymmetric amplification) in the reaction of chalcone or cyclohexanone with diethylzinc catalyzed by the Ni(acac)₂–(–)DAIB system,⁴⁹ and a negative non-linear effect (asymmetric depletion) in the reaction of chalcone with diethylzinc catalyzed by the Cu(OTf)₂–chiral phosphoramidite ligand.⁵⁰

The work reported by Feng and his co-workers is noteworthy. They demonstrated remarkable asymmetric amplification in the 1,4-addition of thioglycolate to chalcones catalyzed by a chiral l-proline-derived N,N'-dioxide-La(OTf)₃ complex (Eq. 5).⁵¹

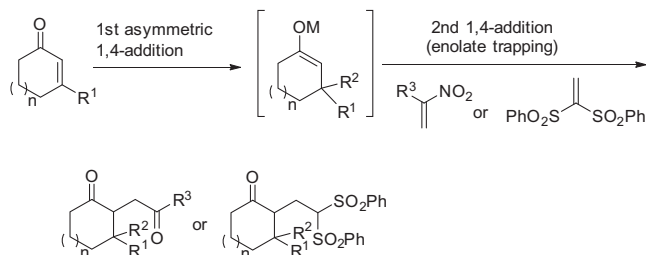




Scheme 14. Catalytic enantioselective annulations via 1,4-addition and aldol cyclization.

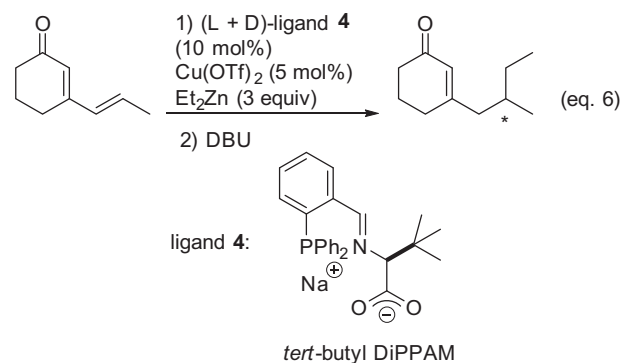


Scheme 15. Trapping of zinc enolate with nitroolefin.



Scheme 16. Trapping of metal enolate with nitroolefins or vinyl *gem*-disulfones.

Alexakis, Crévisy, and Mauduit reported asymmetric amplification in the enantioselective Cu/DiPPAM-catalyzed 1,6- and 1,4-additions of diethylzinc to (di)enones (Eq. 6).^{52,53}



Switching of enantioselectivity⁵⁴

A variety of compounds showing physiological activity, such as pharmaceuticals, pesticides, and perfumes have optically active moieties. In these compounds, one enantiomer often exhibits favorable physiological activity, while the other may exhibit no activity, inhibit favorable physiological activity, or even exhibit unfavorable physiological activity. Therefore, the development of a synthetic methodology that selectively produces one enantiomer with high ee is a very important and challenging topic of research. However, when amino acids or carbohydrates are to be used as a substrate or ligand, only one enantiomer of the product can be obtained. One of the solutions to overcome this limitation is reversal of enantioselectivity, which denotes the selective formation of both enantiomers from the same chiral origin. Reversal of enantioselectivity was observed by controlling the nature of the substituent in the catalyst and/or substrate, changing the central metal, or controlling the solvent, temperature, and additives. In asymmetric 1,4-addition, too, several unique methods to switch the stereochemistry of the products have been proposed. These reports are significant not only in terms of synthetic usefulness but also in providing relevant mechanistic information. In this chapter, we will introduce some recent examples in this regard.

In 2009, Sakaguchi and his co-workers reported complete reversal of enantioselectivity in the Cu-catalyzed 1,4-addition of dialkylzinc to cyclic enone, in the presence of chiral azolium compound.^{21,55,56} Reversal of enantioselectivity was observed by changing the copper species. That is, 1,4-addition of diethylzinc to 2-cyclohexen-1-one catalyzed by Cu(OTf)₂ combined with an azolium salt derived from (*S*)-leucinol produced the (*S*)-1,4-adduct in up to 81% ee, while the combination of the same ligand with Cu(acac)₂ afforded the (*R*)-1,4-adduct in up to 82% ee. The same authors reported that the ee of each product was increased up to >99.5% and 86% by optimization of the substituents in the azolium salt and the type of dialkylzinc (Scheme 20).

In 2012, the same authors reported that both Cu(OTf)₂/CH₂-bridged ligand **5** and Cu(acac)₂/(CH₂)₂-bridged ligand **6** combinations afforded the same enantiomer in up to 99% ee. On the other hand, the Cu(OTf)₂/**6** and Cu(acac)₂/**5** combinations afforded the opposite enantiomer in up to 84% ee (Scheme 21).⁵⁷

Sakaguchi speculated that in the Cu(OTf)₂/**5** catalytic system, the anionic amidate/NHC-Cu species **A** might be involved, whereas the 1,4-addition catalyzed by Cu(acac)₂/**5** may proceed through model **B**, where the anionic acetylacetonate ligand coordinates to a Cu center (Scheme 22).

Very recently, Sakaguchi reported that a change in the order of addition of the substrate (enone) and diethylzinc caused a reversal of enantioselectivity, as shown in Scheme 23.^{58c}

In 2012, Endo and Shibata reported the multinuclear Cu/Zn complex-catalyzed asymmetric 1,4-addition of organozinc

Table 1

Enantioselective 1,4-addition, followed by trapping of zinc enolate with allyl iodides.

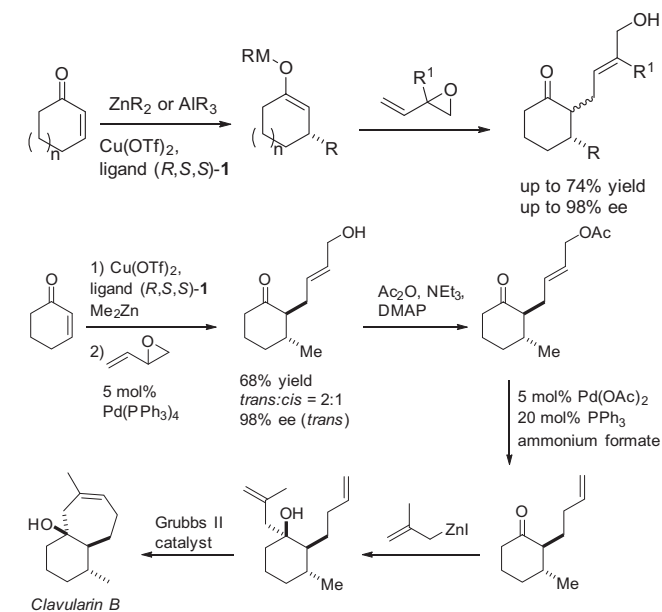
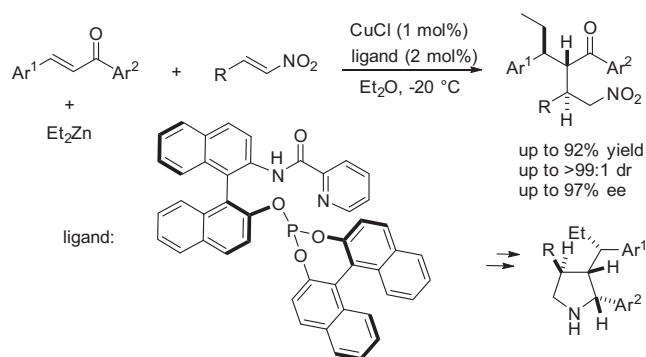
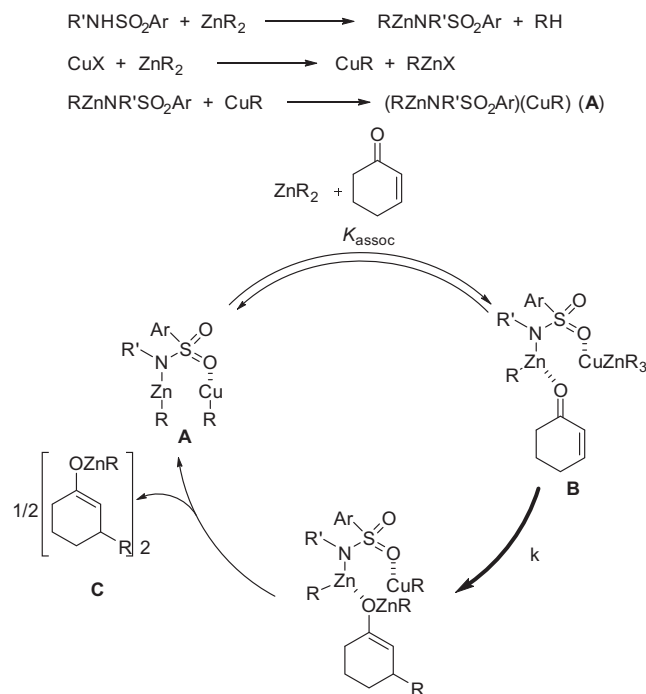
$\text{Cu}(\text{OTf})_2$ (1 mol %)

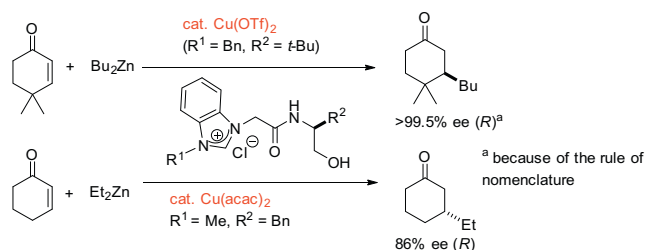
Entry	n	R ¹	R ²	R ³	Time/h	Yield/% ^a	trans:cis	ee of trans
1	1	Et	<i>i</i> -Pr	H	5	76	95:5	99
2	1	Me	<i>i</i> -Pr	H	24	71	93:7	99
3	1	Et	<i>i</i> -Pr	Me	5	85	97:3	99
4	1	Me	<i>i</i> -Pr	Me	24	83	96:4	99
5	0	Et	<i>t</i> -Bu	H	5	44	95:5	97

^a Combined yield of *trans* and *cis* isomers.**Table 2**

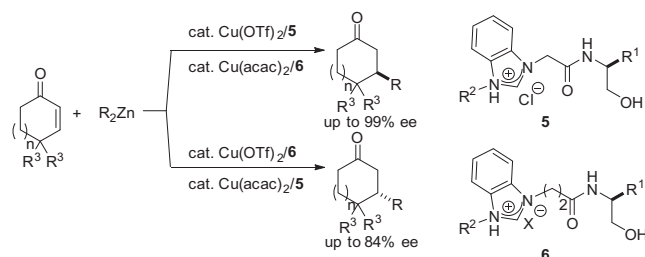
Reaction of dienone 11.

Entry	RM	Ligand	Solvent	12/13	ee%
1	Et ₂ Zn		2-Me-THF	0/100	97
2	EtMgBr		CH ₂ Cl ₂	>99/1	97

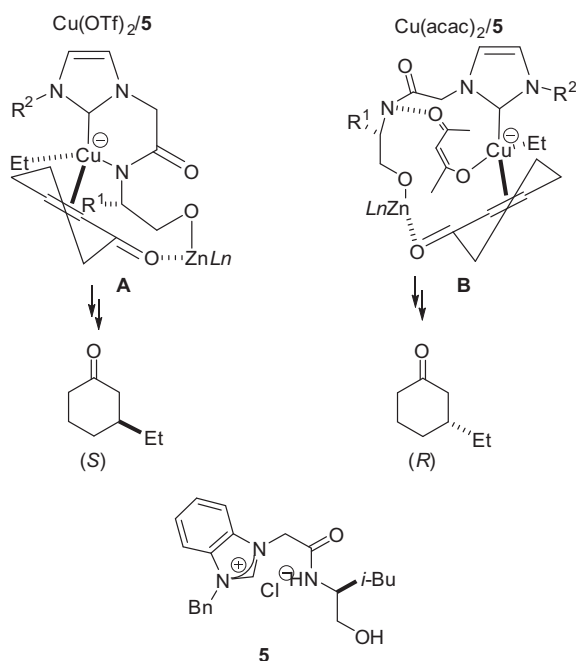
**Scheme 17.** Formal synthesis of Clavularin B.**Scheme 18.** Enantioselective tandem reaction to furnish pyrrolidines.**Scheme 19.** Catalytic cycle of the 1,4-addition of dialkylzincs to α,β -unsaturated ketones.



Scheme 20. Reversal of enantioselectivity by changing copper (II) sources.



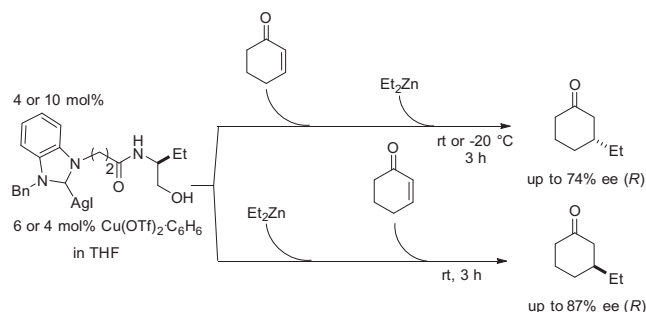
Scheme 21. Reversal of enantioselectivity by modifying the structure of the chiral ligand and metal combination.



Scheme 22. Explanation of the reversal of enantioselectivity by changing copper(II) sources.

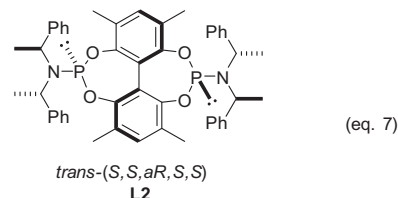
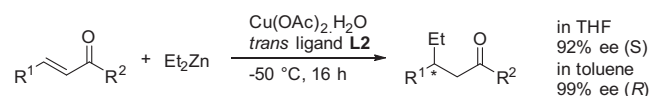
reagents to acyclic and cyclic enones. In this reaction, regioisomeric SPINOL-PHOS ligands gave the opposite enantiomer of the product (Scheme 24).²⁶ The authors proposed the structure of Zn- and Cu/Zn-complexes based on ESI-MS analysis (Scheme 25), although they did not mention the reason for the inversion of stereoselectivity.

Zhang reported that the nature of the solvent dramatically inverted the enantioselectivity.²⁸ That is, the reaction of chalcone with diethylzinc in the presence of 1 mol% of Cu(OAc)₂·H₂O and



Scheme 23. Switching of enantioselectivity by modifying the addition order.

2 mol% of the ligand in THF afforded the 1,4-adduct in 94% yields and with 92% ee (S). On the other hand, the reaction in toluene gave the product in 96% yield and with 99% ee (R) (Eq. 7).

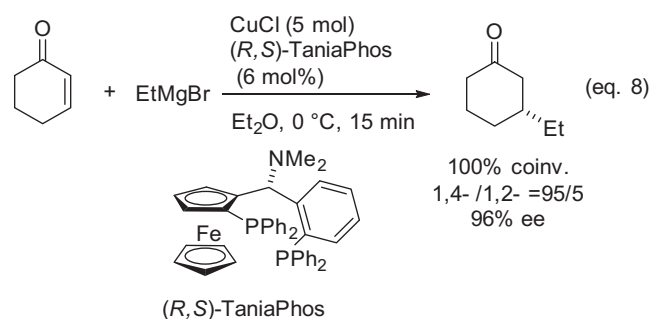


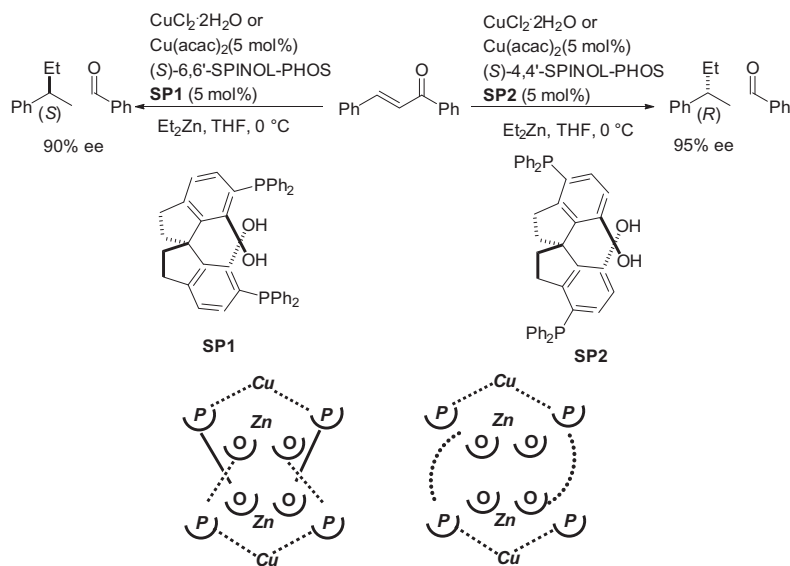
They explained this reversal phenomenon as shown in Scheme 26. The coordination of THF to the metal affects the asymmetric environment (refer to the original paper for more details).

Combination of Grignard reagents with copper

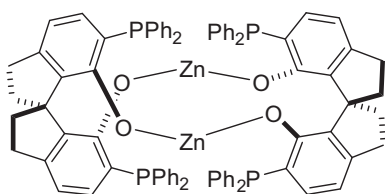
The nucleophile used in copper-catalyzed 1,4-additions is not limited to dialkylzincs; other organometallic reagents such as Grignard reagents and organoaluminum reagents have been also studied.

In 1988, Lippard reported the first enantioselective 1,4-addition of a Grignard reagent to an enone using a catalytic amount of copper amide complex, but the enantioselectivities were low.⁵⁹ Subsequently, many reports on asymmetric 1,4-addition have been published. However, the enantioselectivities were low except in two cases. In 2004, Feringa first reported the highly enantioselective (up to 96% ee) copper-catalyzed asymmetric conjugate addition of Grignard reagents to cyclic enones (Eq. 8).⁶⁰



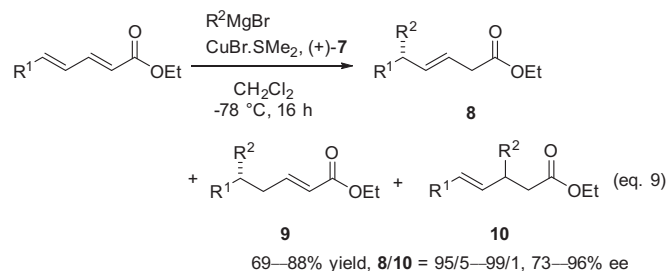


Scheme 24. Concept of multinuclear Cu/Zn complex.

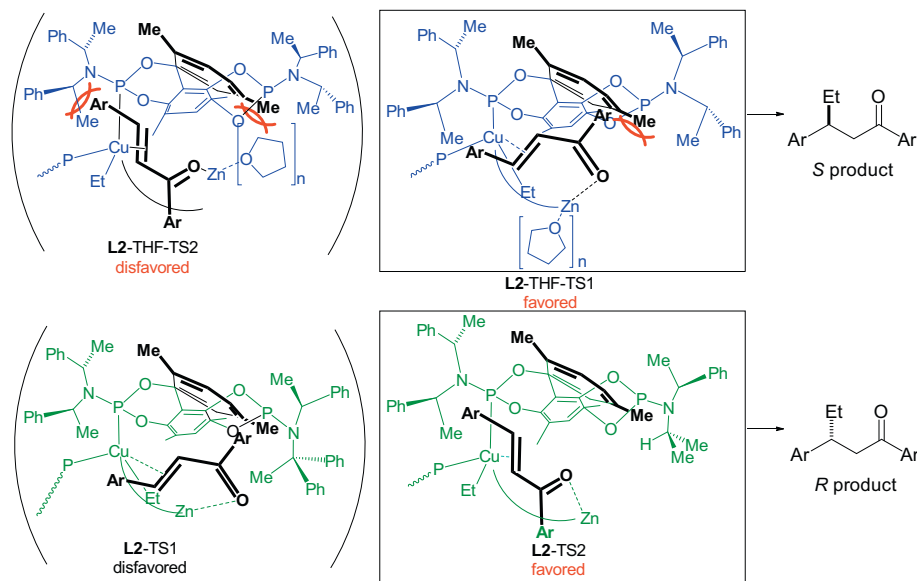


Scheme 25. Proposed structure of Zn/SP1-complex.

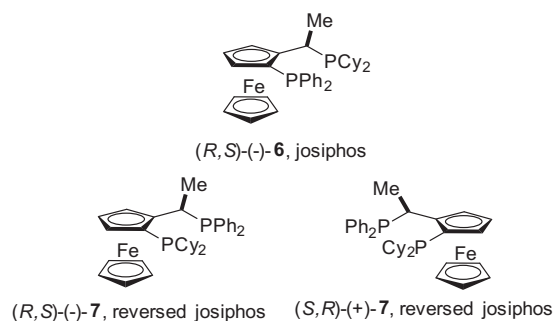
1,6-adduct **8** with high regio- and enantioselectivity. Only a trace amount (<2%) of the 1,4-adduct was formed, and the 1,2-addition product was not detected at all.



In 2008, Minnaard and Feringa reported the catalytic enantioselective 1,6-addition of Grignard reagents to linear dienones (Eq. 9).⁶¹ In the presence of *(R,S)*-(-)-reversed josiphos **7** or *(S,R)*-(+)-reversed josiphos (5.25 mol%) and $\text{CuBr}\cdot\text{SMe}_2$ (5 mol%), the addition of EtMgBr to β,γ -unsaturated 1,6-addition proceeded to afford

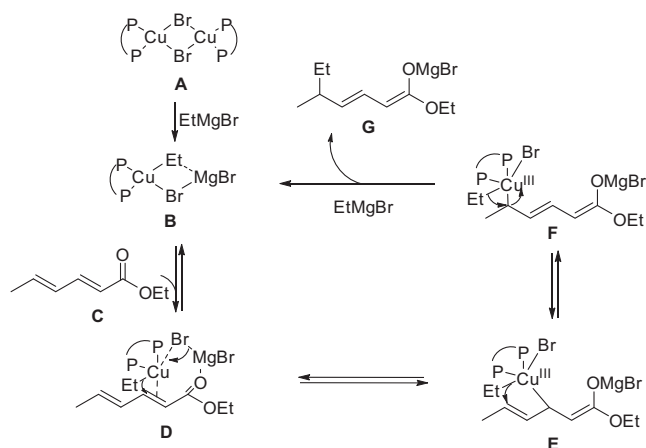


Scheme 26. Reversal of enantioselectivity by choice of solvent.



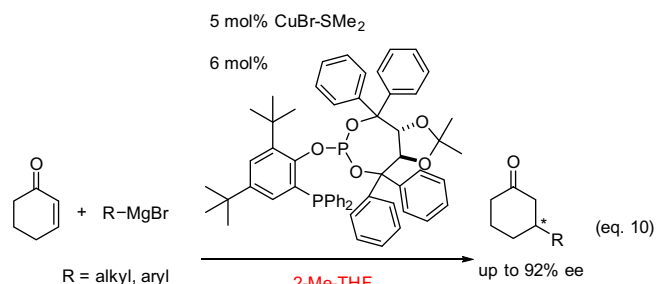
Based on their own mechanistic study⁶² and the theoretical studies by Nakamura⁶³ and Morokuma's⁶⁴ groups, Feringa proposed the mechanism shown in Scheme 27.

The catalytic cycle starts with formation of reactive complex **B** from the dimeric resting state **A** of the catalyst. Intermediate **B** forms a π complex **D** with substrate **C**, followed by the formation of the copper (III) σ complex **E**. Then, **E** undergoes sequential copper migration via σ/π -allylcopper (III) complex **F** to the remote position. The catalytic cycle ends with reductive elimination to form product **G**. The preference for the 1,6-addition product over the 1,4-product is explained by the lower activation energy for the migration of the Cu complex as compared to that for alkyl addition at the 4-position, because this addition disturbs the conjugate system.



Scheme 27. Proposed catalytic cycle for the asymmetric 1,6-addition of EtMgBr to ethyl sorbate.

Schmalz reported the enantioselective copper-catalyzed 1,4-addition of Grignard reagents to 2-cyclohexen-1-one using a Taddol-derived phosphine-phosphite ligand in 2-methyl-THF as a solvent (Eq. 10).⁶⁵ **

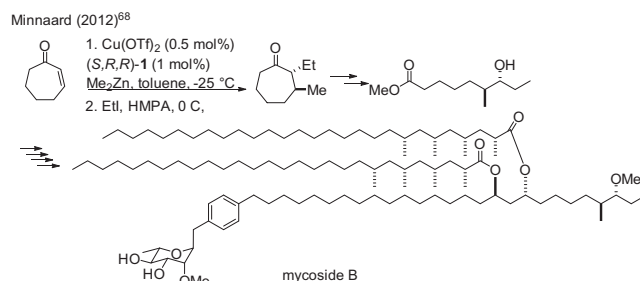


Thaler and Knochel reported the copper-catalyzed asymmetric 1,4-addition of magnesium organometallic reagents to provide chiral molecules, as shown in Scheme 28.⁶⁶

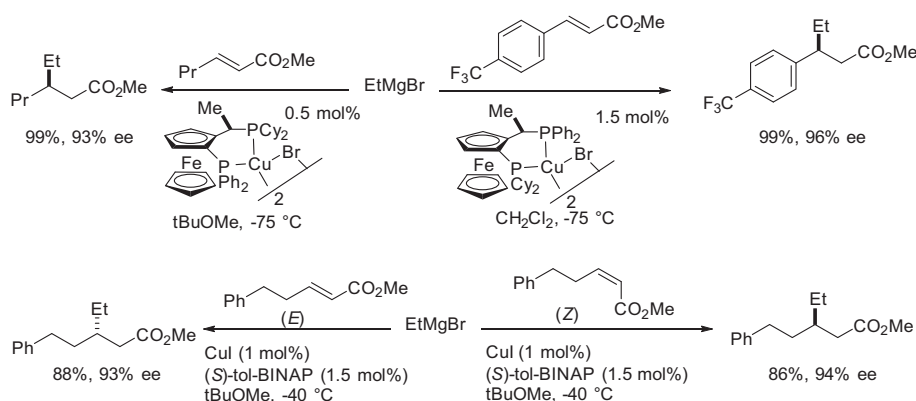
In 2012, Guénée, Mauduit, and Alexakis reported the formation of quaternary stereogenic centers by NHC-Cu-catalyzed asymmetric 1,4-addition with Grignard reagents on polyconjugated cyclic enones. It is noteworthy that the reaction of **11** with diethylzinc exclusively furnished the 1,6-addition product **13**.⁶⁷

Synthetic application to natural products

In this chapter, we will demonstrate the total synthesis of natural products via a catalytic asymmetric conjugate reaction. Minnaard reported the total synthesis of phenolic glycolipid mycoside B and glycosylated hydroxybenzoic acid methyl ester HBAD-I, virulence markers of *Mycobacterium tuberculosis* (Scheme 29).⁶⁸



Scheme 29. Synthesis of mycoside B.



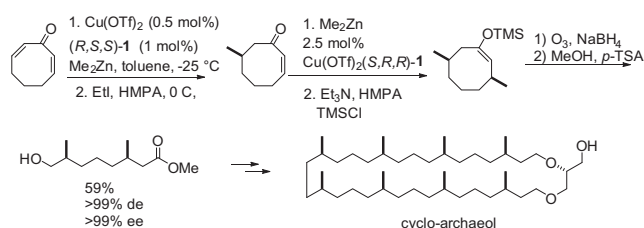
Scheme 28. Copper-catalyzed 1,4-addition of organomagnesium reagents.

The same author reported the total synthesis of cyclo-archaeol (Scheme 30).⁶⁹

Finally, we would like to show two recent reports of the enantioselective total synthesis of natural products, in which catalytic asymmetric 1,4-addition was used as the key step to generate the chiral center. Luo reported the synthesis of amphilectane and serrulatane diterpenoids. Asymmetric 1,4-addition was used for the synthesis of erogorgiaene (Scheme 31).⁷⁰

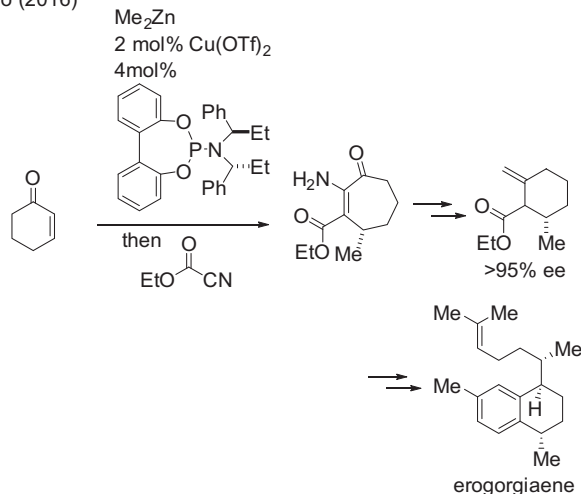
Baran⁷¹ accomplished a 11-step synthesis of (–)-maoecrystal V starting from an enantioselective 1,4-addition modified by Schmalz's catalyst and solvent system (Scheme 32).⁶⁵

Schouten and Minnaard (2013)⁶⁹



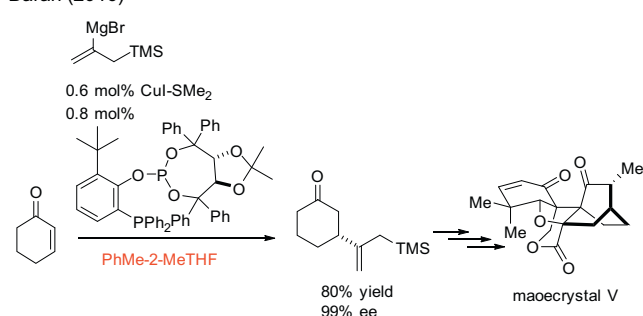
Scheme 30. Synthesis of cyclo-archaeol.

Luo (2016)⁷⁰



Scheme 31. Synthesis of erogorgiaene.

Baran (2016)⁷¹



Scheme 32. synthesis of maoecrystal V.

Conclusion

We have summarized the recent progress in catalytic asymmetric 1,4-additions. Efforts for the development of chiral ligands and chiral catalysts toward 100% yield and 100% ee with a high TON and TOF will be continued for a variety of Michael acceptors. Attempts to extend catalytic asymmetric 1,4-addition to the synthesis of complex natural products have just been undertaken. In the early stages of research, dialkylzincs were the first choice for organometallic nucleophiles; however, Grignard reagents and other organometallic compounds such as organoborons^{72,73} have recently been used in copper-catalyzed 1,4-addition. Note that space constraints prevented us from including some significant reports on Rh, Ir, Pd-catalyzed^{74–87} and organocatalytic^{88,89} asymmetric 1,4-addition of organoboron reagents. More details of these studies can be found in the cited references.

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